Note 5

Question 1

a. `ls`

- **Definition:**
  - Lists files and directories.
- **Usage/Formula:**
  - ls [options] [directory].
- **Example:**
  - [directory] ls — list the current directory's contents. `ls -a` — include hidden files. `ls -lh` — show a detailed list with human-readable file sizes.

b. `pwd`

- **Definition:**
  - Prints the absolute path of current working directory.
- **Usage/Formula:**
  - pwd
- **Example:**
  - pwd — might display /home/shanto/marstenHouse.

c. `cd`

- **Definition:**
  - Changes your current working directory.
- **Usage/Formula:**
  - cd [directory]
- **Example:**
  - cd hallway — enter the hallway directory. `cd ..` — move up one directory. `cd ~` — go to your home directory.

Question 2

a. What is a variable? A variable is a named place that stores a value, such as text or a number.

b. How do I use a variable? assign a value using = sign `name="Shanto"` `echo "$name"` Output: Shanto

c. What is an environment variable? A variable passed to programs started from the shell. For example, HOME stores your home directory's path. `echo "$HOME"`

d. What is a user defined variable? A variable we create ourself. `course="Linux"` `echo "$course"`

e. What is the root directory? The top directory of the Linux filesystem, written as /. All other directories are beneath it. `cd ..`

f. What does “Parent Directory” mean? The directory directly above another directory. .. refers to the parent. `cd ..`

g. What does “Current working directory” mean? The directory you are currently working in. Use `pwd` to display its path.

h. What is an absolute path? Include an example A complete path starting from the root directory /. Example: `/home/shanto/marstenHouse/hallway/.key2`

i. What is a relative path? Include an example What is a relative path? A path starting from your current working directory. Example: `hallway/.key2` refers to that key when you are inside marstenHouse.

j. What is the difference between “Your home directory” and “The home directory”? Your home directory is your personal folder, such as `/home/shanto` You can refer to it as ~ or $HOME. The /home directory usually contains the personal home folders of regular users, such as `/home/shanto` and `/home/student`.